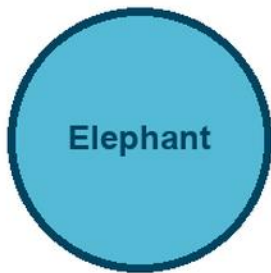


KITABU QUEST

Form III

PHYSICS



Elephant

Cover scene

students exploring physics with apparatus and an elephant mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

II

Title Page

KITABU QUEST Form III Physics

Tanzania TIE-BASIC learner book

Mascot: elephant

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Form III learners study Physics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use measurements, diagrams, models, and step-by-step reasoning about systems and forces.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Current Electricity And Circuit Analysis
2. Electromagnetism And Motors
3. Electronics And Semiconductors Basics
4. Pressure Fluids And Hydraulics
5. Circular Motion And Gravitation Basics
6. Radioactivity And Nuclear Safety
7. Measurement Errors And Data
8. Form III Physics Practical Review

As you finish each topic, return to the success criteria and correct one answer before moving on.

Current Electricity And Circuit Analysis: Lesson Opener

Learning focus: Current Electricity And Circuit Analysis

Country link: a safe Tanzania observation, model, data table, or investigation for Current Electricity And Circuit Analysis.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain current electricity and circuit analysis using evidence

Inquiry questions:

- How can current electricity and circuit analysis help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Current Electricity And Circuit Analysis: Learn

Current Electricity And Circuit Analysis explains moving electric charge in circuits.

Core idea:

- A complete circuit is needed for current to flow.
- Cells, wires, switches, and bulbs can show current effects safely.
- Electricity can produce light, heat, sound, movement, and magnetism.
- Safety matters: never experiment with mains electricity.

Key words:

- Current Electricity And Circuit Analysis
- Current
- Electricity
- Circuit
- Analysis
- concept
- observation
- evidence

Current Electricity And Circuit Analysis: Worked Example

Circuit model for Current Electricity And Circuit Analysis: Light a bulb safely.

1. Use a cell, two wires, a switch, and a small bulb.
2. Connect a complete circuit.
3. Close the switch and observe the bulb.
4. Open the switch and observe again.

Conclusion: current flows only when the circuit is complete.

Current Electricity And Circuit Analysis: Vocabulary And Study Skill

Use these words and study moves before you practise Current Electricity And Circuit Analysis. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Current Electricity And Circuit Analysis
- Current
- Electricity
- Circuit
- Analysis
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Current Electricity And Circuit Analysis without a clear example, method, or evidence.

Current Electricity And Circuit Analysis: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Current Electricity And Circuit Analysis.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Current Electricity And Circuit Analysis: Practice And Correction

Practice for Current Electricity And Circuit Analysis:

Main outcome: investigate and explain current electricity and circuit analysis using evidence.

1. Draw a complete circuit with a cell, switch, and bulb.
2. Explain why a broken circuit stops current.
3. List two electrical safety rules.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Current Electricity And Circuit Analysis: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Current Electricity And Circuit Analysis.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Current Electricity And Circuit Analysis: Outcome Check

Outcome check for Current Electricity And Circuit Analysis: Investigate and explain current electricity and circuit analysis using evidence.

1. Draw a complete circuit.
2. Explain why a switch can stop current.
3. Write one electrical safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electromagnetism And Motors: Lesson Opener

Learning focus: Electromagnetism And Motors

Country link: a safe Tanzania observation, model, data table, or investigation for Electromagnetism And Motors.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain electromagnetism and motors using evidence

Inquiry questions:

- How can electromagnetism and motors help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electromagnetism And Motors: Learn

Electromagnetism And Motors explains magnetic and non-magnetic materials.

Core idea:

- Magnets attract some materials, especially iron and steel.
- A magnet has north and south poles.
- Like poles repel and unlike poles attract.
- Magnetic force can act without touching the object.

Key words:

- Electromagnetism And Motors
- Electromagnetism
- Motors
- concept
- observation
- evidence
- safety
- conclusion

Electromagnetism And Motors: Worked Example

Investigation model for Electromagnetism And Motors: Test materials with a magnet.

1. Collect a nail, coin, plastic ruler, paper clip, and wood piece.
2. Bring a magnet near each object.
3. Record attracted or not attracted.
4. Group the materials as magnetic or non-magnetic.

Conclusion: iron and steel objects are usually attracted by magnets.

Electromagnetism And Motors: Vocabulary And Study Skill

Use these words and study moves before you practise Electromagnetism And Motors. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electromagnetism And Motors
- Electromagnetism
- Motors
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electromagnetism And Motors without a clear example, method, or evidence.

Electromagnetism And Motors: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Electromagnetism And Motors.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electromagnetism And Motors: Practice And Correction

Practice for Electromagnetism And Motors:

Main outcome: investigate and explain electromagnetism and motors using evidence.

1. Test five safe materials with a magnet.
2. Record attracted/not attracted in a table.
3. State what happens when like poles are brought together.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electromagnetism And Motors: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electromagnetism And Motors.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electromagnetism And Motors: Outcome Check

Outcome check for Electromagnetism And Motors: Investigate and explain electromagnetism and motors using evidence.

1. Name two magnetic materials.
2. State what happens with like poles.
3. Describe a fair magnet test.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electronics And Semiconductors Basics: Lesson Opener

Learning focus: Electronics And Semiconductors Basics

Country link: a safe Tanzania observation, model, data table, or investigation for Electronics And Semiconductors Basics.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain electronics and semiconductors basics using evidence

Inquiry questions:

- How can electronics and semiconductors basics help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electronics And Semiconductors Basics: Learn

Electronics And Semiconductors Basics is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Electronics And Semiconductors Basics
- Electronics
- Semiconductors
- Basics
- concept
- observation
- evidence
- safety

Electronics And Semiconductors Basics: Worked Example

Investigation model for Electronics And Semiconductors Basics: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Electronics And Semiconductors Basics: Vocabulary And Study Skill

Use these words and study moves before you practise Electronics And Semiconductors Basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electronics And Semiconductors Basics
- Electronics
- Semiconductors
- Basics
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electronics And Semiconductors Basics without a clear example, method, or evidence.

Electronics And Semiconductors Basics: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Electronics And Semiconductors Basics.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electronics And Semiconductors Basics: Practice And Correction

Practice for Electronics And Semiconductors Basics:

Main outcome: investigate and explain electronics and semiconductors basics using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electronics And Semiconductors Basics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electronics And Semiconductors Basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electronics And Semiconductors Basics: Outcome Check

Outcome check for Electronics And Semiconductors Basics: Investigate and explain electronics and semiconductors basics using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Pressure Fluids And Hydraulics: Lesson Opener

Learning focus: Pressure Fluids And Hydraulics

Country link: a safe Tanzania observation, model, data table, or investigation for Pressure Fluids And Hydraulics.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain pressure fluids and hydraulics using evidence

Inquiry questions:

- How can pressure fluids and hydraulics help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Pressure Fluids And Hydraulics: Learn

Pressure Fluids And Hydraulics is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Pressure Fluids And Hydraulics
- Pressure
- Fluids
- Hydraulics
- concent

Pressure Fluids And Hydraulics: Worked Example

Investigation model for Pressure Fluids And Hydraulics: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Pressure Fluids And Hydraulics: Vocabulary And Study Skill

Use these words and study moves before you practise Pressure Fluids And Hydraulics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Pressure Fluids And Hydraulics
- Pressure
- Fluids
- Hydraulics
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Pressure Fluids And Hydraulics without a clear example, method, or evidence.

Pressure Fluids And Hydraulics: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Pressure Fluids And Hydraulics.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Pressure Fluids And Hydraulics: Practice And Correction

Practice for Pressure Fluids And Hydraulics:

Main outcome: investigate and explain pressure fluids and hydraulics using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Pressure Fluids And Hydraulics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Pressure Fluids And Hydraulics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Pressure Fluids And Hydraulics: Outcome Check

Outcome check for Pressure Fluids And Hydraulics: Investigate and explain pressure fluids and hydraulics using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Circular Motion And Gravitation Basics: Lesson Opener

Learning focus: Circular Motion And Gravitation Basics

Country link: a safe Tanzania observation, model, data table, or investigation for Circular Motion And Gravitation Basics.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain circular motion and gravitation basics using evidence

Inquiry questions:

- How can circular motion and gravitation basics help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Circular Motion And Gravitation Basics: Learn

Circular Motion And Gravitation Basics explains how position changes with time.

Core idea:

- Motion means an object changes position compared with a reference point.
- Distance tells how far an object moves.
- Time tells how long the motion takes.
- Speed compares distance and time.
- A distance-time table or graph can show whether motion is slow, fast, or changing.

Key words:

- Circular Motion And Gravitation Basics
- Circular
- Motion
- Gravitation
- Basics
- concept
- observation

Circular Motion And Gravitation Basics: Worked Example

Worked example for Circular Motion And Gravitation Basics: A cart moves 12 metres in 4 seconds. Find its speed.

1. Speed = distance / time.
2. Distance = 12 m and time = 4 s.
3. $12 / 4 = 3$.

Answer: the cart moves at 3 m/s.

Circular Motion And Gravitation Basics: Vocabulary And Study Skill

Use these words and study moves before you practise Circular Motion And Gravitation Basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Circular Motion And Gravitation Basics
- Circular
- Motion
- Gravitation
- Basics
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Circular Motion And Gravitation Basics without a clear example, method, or evidence.

Circular Motion And Gravitation Basics: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Circular Motion And Gravitation Basics.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Circular Motion And Gravitation Basics: Practice And Correction

Practice for Circular Motion And Gravitation Basics:

Main outcome: investigate and explain circular motion and gravitation basics using evidence.

1. A bicycle moves 30 m in 6 s. Calculate its speed.
2. Draw a two-row table for distance and time.
3. Explain what happens to speed if the same distance takes more time.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Circular Motion And Gravitation Basics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Circular Motion And Gravitation Basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Circular Motion And Gravitation Basics: Outcome Check

Outcome check for Circular Motion And Gravitation Basics: Investigate and explain circular motion and gravitation basics using evidence.

1. Calculate the speed of an object that moves 20 m in 5 s.
2. State the formula used.
3. Write the unit.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Radioactivity And Nuclear Safety: Lesson Opener

Learning focus: Radioactivity And Nuclear Safety

Country link: a safe Tanzania observation, model, data table, or investigation for Radioactivity And Nuclear Safety.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain radioactivity and nuclear safety using evidence

Inquiry questions:

- How can radioactivity and nuclear safety help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Radioactivity And Nuclear Safety: Learn

Radioactivity And Nuclear Safety explains radioactivity as the process where unstable nuclei emit radiation to become more stable.

Core idea:

- Alpha, beta, and gamma radiation differ in charge, mass, penetration, and shielding.
- Half-life is the time taken for half of a radioactive sample or count rate to remain.
- Radioactivity can be useful in medicine, industry, agriculture, dating, and research when controlled safely.
- Radiation can also damage living tissue, so distance, shielding, time limits, and trained supervision are essential.
- Learners should interpret simple count-rate data and avoid unsafe experiments.

Key words:

- Radioactivity And Nuclear Safety
- Radioactivity
- Nuclear
- Safety
- concept
- observation
- evidence
- safety

Radioactivity And Nuclear Safety: Worked Example

Worked example for Radioactivity And Nuclear Safety: Use half-life data.

A sample has a count rate of 80 counts per minute. Its half-life is 5 minutes.

1. After 5 minutes, half remains: 40 counts per minute.
2. After 10 minutes, half of 40 remains: 20 counts per minute.
3. After 15 minutes, half of 20 remains: 10 counts per minute.
4. Shielding and distance must be controlled by trained adults.

Answer: after 15 minutes, the count rate is 10 counts per minute.

Radioactivity And Nuclear Safety: Vocabulary And Study Skill

Use these words and study moves before you practise Radioactivity And Nuclear Safety. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Radioactivity And Nuclear Safety
- Radioactivity
- Nuclear
- Safety
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Radioactivity And Nuclear Safety without a clear example, method, or evidence.

Radioactivity And Nuclear Safety: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Radioactivity And Nuclear Safety.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Radioactivity And Nuclear Safety: Practice And Correction

Practice for Radioactivity And Nuclear Safety:

Main outcome: investigate and explain radioactivity and nuclear safety using evidence.

1. Compare alpha, beta, and gamma radiation in a table.
2. Solve one half-life count-rate question.
3. Write two radiation safety rules involving time, distance, or shielding.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Radioactivity And Nuclear Safety: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Radioactivity And Nuclear Safety.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Radioactivity And Nuclear Safety: Outcome Check

Outcome check for Radioactivity And Nuclear Safety: Investigate and explain radioactivity and nuclear safety using evidence.

1. Distinguish alpha, beta, and gamma radiation.
2. Use half-life in one simple calculation.
3. Write one radiation safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Measurement Errors And Data: Lesson Opener

Learning focus: Measurement Errors And Data

Country link: a safe Tanzania observation, model, data table, or investigation for Measurement Errors And Data.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain measurement errors and data using evidence

Inquiry questions:

- How can measurement errors and data help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Measurement Errors And Data: Learn

Measurement Errors And Data begins with safe behaviour and accurate measurement.

Core idea:

- Physics uses measurements such as length, mass, time, temperature, and volume.
- A measurement is useful only when the number and unit are both written.
- Safety rules protect learners, equipment, and results.
- Read scales at eye level and record values immediately.

Local link: in Tanzania, careful measurement helps in health, building, farming, weather records, transport, and school laboratory work.

Key words:

- Measurement Errors And Data
- Measurement
- Errors
- Data
- concept
- observation
- evidence
- safety

Measurement Errors And Data: Worked Example

Investigation model for Measurement Errors And Data: Measure the length of a desk safely.

1. Choose a metre rule or tape measure.
2. Place zero at one end of the desk.
3. Read the scale at eye level.
4. Record the value with the unit, for example 120 cm.
5. Repeat once and compare readings.

Conclusion: a useful measurement includes a number, a unit, and a careful method.

Measurement Errors And Data: Vocabulary And Study Skill

Use these words and study moves before you practise Measurement Errors And Data. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Measurement Errors And Data
- Measurement
- Errors
- Data
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Measurement Errors And Data without a clear example, method, or evidence.

Measurement Errors And Data: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Measurement Errors And Data.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Measurement Errors And Data: Practice And Correction

Practice for Measurement Errors And Data:

Main outcome: investigate and explain measurement errors and data using evidence.

1. Name three laboratory safety rules.
2. Measure one classroom object and record the value with a unit.
3. Explain one error that can make a measurement unreliable.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Measurement Errors And Data: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Measurement Errors And Data.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Measurement Errors And Data: Outcome Check

Outcome check for Measurement Errors And Data: Investigate and explain measurement errors and data using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form III Physics Practical Review: Lesson Opener

Learning focus: Form III Physics Practical Review

Country link: a safe Tanzania observation, model, data table, or investigation for Form III Physics Practical Review.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain form iii physics practical review using evidence

Inquiry questions:

- How can form iii physics practical review help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form III Physics Practical Review: Learn

Form III Physics Practical Review is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Form III Physics Practical Review
- Form
- Physics
- Practical
- Review
- concept
- observation
- evidence

Form III Physics Practical Review: Worked Example

Investigation model for Form III Physics Practical Review: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Form III Physics Practical Review: Vocabulary And Study Skill

Use these words and study moves before you practise Form III Physics Practical Review. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Form III Physics Practical Review
- Form
- Physics
- Practical
- Review
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Form III Physics Practical Review without a clear example, method, or evidence.

Form III Physics Practical Review: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Form III Physics Practical Review.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Form III Physics Practical Review: Practice And Correction

Practice for Form III Physics Practical Review:

Main outcome: investigate and explain form iii physics practical review using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Form III Physics Practical Review: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Form III Physics Practical Review.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form III Physics Practical Review: Outcome Check

Outcome check for Form III Physics Practical Review: Investigate and explain form iii physics practical review using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Current Electricity And Circuit Analysis: Review Clinic 1

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Current Electricity And Circuit Analysis that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Electromagnetism And Motors: Review Clinic 2

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Electromagnetism And Motors that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Electronics And Semiconductors Basics: Review Clinic 3

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Electronics And Semiconductors Basics that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Pressure Fluids And Hydraulics: Review Clinic 4

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Pressure Fluids And Hydraulics that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Current Electricity And Circuit Analysis: Application Workshop 1

Application workshop 1 for Current Electricity And Circuit Analysis.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Electromagnetism And Motors: Application Workshop 2

Application workshop 2 for Electromagnetism And Motors.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Electronics And Semiconductors Basics: Application Workshop 3

Application workshop 3 for Electronics And Semiconductors Basics.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Pressure Fluids And Hydraulics: Application Workshop 4

Application workshop 4 for Pressure Fluids And Hydraulics.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Circular Motion And Gravitation Basics: Application Workshop 5

Application workshop 5 for Circular Motion And Gravitation Basics.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Radioactivity And Nuclear Safety: Application Workshop 6

Application workshop 6 for Radioactivity And Nuclear Safety.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Measurement Errors And Data: Application Workshop 7

Application workshop 7 for Measurement Errors And Data.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Form III Physics Practical Review: Application Workshop 8

Application workshop 8 for Form III Physics Practical Review.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Current Electricity And Circuit Analysis: Application Workshop 9

Application workshop 9 for Current Electricity And Circuit Analysis.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Electromagnetism And Motors: Application Workshop 10

Application workshop 10 for Electromagnetism And Motors.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Glossary

Use this glossary to revise important words in Physics.

- Current Electricity And Circuit Analysis: Explain this term using an example from the book, your school, or your community.
- Electromagnetism And Motors: Explain this term using an example from the book, your school, or your community.
- Electronics And Semiconductors Basics: Explain this term using an example from the book, your school, or your community.
- Pressure Fluids And Hydraulics: Explain this term using an example from the book, your school, or your community.
- Circular Motion And Gravitation Basics: Explain this term using an example from the book, your school, or your community.
- Radioactivity And Nuclear Safety: Explain this term using an example from the book, your school, or your community.
- Measurement Errors And Data: Explain this term using an example from the book, your school, or your community.
- Form III Physics Practical Review: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Physics answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence.

Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Physics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.